

Automated IPAM & BoQ Project:
Implementing Multi-VLAN DHCP with Active Port Security

NAMA : KEYSA ANNAJWA
KELAS : XI SMK TKJ
TUJUAN : Melatih integrasi data inventaris jaringan (BoQ), segmentasi keamanan (VLAN), dan otomatis pemetaan IP Address
TOOLS/RUMUS : =COUNT(), =SUM(), VLOOKUP, DATA VALIDATION, Cisco Packet Tracer (CT),

1. HOST SPECIFICATION & OS DISTRIBUTION MATRIX

DEVICE DATA ALL						
NO	Device ID	Device Name	Brand	Model	OS Version	Location (Floor / 2)
1	PC-1	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL
2	PC-2	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL
3	PC-3	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL
4	PC-4	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL
5	PC-5	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL
6	PC-6	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL
7	PC-7	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL
8	PC-8	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL
9	PC-9	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL
10	PC-10	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL
11	PC-11	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL
12	PC-12	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL
13	PC-13	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL
14	PC-14	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL
15	PC-15	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL
16	PC-16	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL
17	PC-17	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL
18	PC-18	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL
19	PC-19	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL
20	PC-20	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL & TECH
21	PC-21	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL & TECH
22	PC-22	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL & TECH
23	PC-23	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL & TECH
24	PC-24	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL & TECH
25	PC-25	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL & TECH
26	PC-26	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL & TECH
27	PC-27	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL & TECH
28	PC-28	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL & TECH
29	PC-29	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL & TECH
30	PC-30	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL & TECH
31	PC-31	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL & TECH
32	PC-32	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL & TECH
33	PC-33	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL & TECH
34	PC-34	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL & TECH
35	PC-35	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL & TECH
36	PC-36	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL & TECH
37	PC-37	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL & TECH
38	PC-38	PC	Cisco/Broad	Intel Core i5-4570/4GB	Windows 10 Pro 64 bit	SG001 OPERATIONAL & TECH
39	Switch F-A	SWITCH	ARX8016	Arista 7260-48G-4S-16G-2S	EOS 4.25.5	SG001 OPERATIONAL
40	Switch F-B	SWITCH	ARX8016	Arista 7260-48G-4S-16G-2S	EOS 4.25.5	SG001 OPERATIONAL & TECH
41	Router-U	ROUTER	ARX8016	Arista 7260-48G-4S-16G-2S	EOS 4.25.5	SG001 OPERATIONAL & TECH
42	Switch-U	SWITCH	ARX8016	Arista 7260-48G-4S-16G-2S	EOS 4.25.5	SG001 OPERATIONAL & TECH

2. AUTOMATED PORT & IP MAPPING MATRIX (FLOOR 1 & 2)

Automated Port & IP Mapping						
NO	Device ID	Device Name	VLAN ID	IP Address	Interface	Status
1	PC-1	CS	VLAN 10	192.168.10.2	FastEthernet0/1	ACTIVE
2	PC-2	CS	VLAN 10	192.168.10.3	FastEthernet0/2	ACTIVE
3	PC-3	CS	VLAN 10	192.168.10.4	FastEthernet0/3	ACTIVE
4	PC-4	CS	VLAN 10	192.168.10.5	FastEthernet0/4	ACTIVE
5	PC-5	CS	VLAN 10	192.168.10.6	FastEthernet0/5	ACTIVE
6	PC-6	BSGP/PC0001	VLAN 20	192.168.20.2	FastEthernet0/6	ACTIVE
7	PC-7	BSGP/PC0001	VLAN 20	192.168.20.3	FastEthernet0/7	ACTIVE
8	PC-8	BSGP/PC0001	VLAN 20	192.168.20.4	FastEthernet0/8	ACTIVE
9	PC-9	BSGP/PC0001	VLAN 20	192.168.20.5	FastEthernet0/9	ACTIVE
10	PC-10	BSGP/PC0001	VLAN 20	192.168.20.6	FastEthernet0/10	ACTIVE
11	PC-11	FRANCA	VLAN 30	192.168.30.2	FastEthernet0/11	ACTIVE
12	PC-12	FRANCA	VLAN 30	192.168.30.3	FastEthernet0/12	ACTIVE
13	PC-13	FRANCA	VLAN 30	192.168.30.4	FastEthernet0/13	ACTIVE
14	PC-14	FRANCA	VLAN 30	192.168.30.5	FastEthernet0/14	ACTIVE
15	PC-15	FRANCA	VLAN 30	192.168.30.6	FastEthernet0/15	ACTIVE
16	PC-16	FRANCA	VLAN 30	192.168.30.7	FastEthernet0/16	ACTIVE
17	PC-17	FRANCA	VLAN 30	192.168.30.8	FastEthernet0/17	ACTIVE
18	PC-18	FRANCA	VLAN 30	192.168.30.9	FastEthernet0/18	ACTIVE
19	PC-19	FRANCA	VLAN 30	192.168.30.10	FastEthernet0/19	ACTIVE
20	PC-20	FRANCA	VLAN 30	192.168.30.11	FastEthernet0/20	ACTIVE
21	PC-21	FRANCA	VLAN 40	192.168.40.2	FastEthernet0/21	ACTIVE
22	PC-22	FRANCA	VLAN 40	192.168.40.3	FastEthernet0/22	ACTIVE
23	PC-23	FRANCA	VLAN 40	192.168.40.4	FastEthernet0/23	ACTIVE
24	PC-24	FRANCA	VLAN 40	192.168.40.5	FastEthernet0/24	ACTIVE
25	PC-25	FRANCA	VLAN 40	192.168.40.6	FastEthernet0/25	ACTIVE
26	PC-26	FRANCA	VLAN 40	192.168.40.7	FastEthernet0/26	ACTIVE
27	PC-27	FRANCA	VLAN 40	192.168.40.8	FastEthernet0/27	ACTIVE
28	PC-28	FRANCA	VLAN 40	192.168.40.9	FastEthernet0/28	ACTIVE
29	PC-29	FRANCA	VLAN 40	192.168.40.10	FastEthernet0/29	ACTIVE
30	PC-30	FRANCA	VLAN 40	192.168.40.11	FastEthernet0/30	ACTIVE
31	PC-31	FRANCA	VLAN 40	192.168.40.12	FastEthernet0/31	ACTIVE
32	PC-32	FRANCA	VLAN 40	192.168.40.13	FastEthernet0/32	ACTIVE
33	PC-33	FRANCA	VLAN 40	192.168.40.14	FastEthernet0/33	ACTIVE
34	PC-34	FRANCA	VLAN 40	192.168.40.15	FastEthernet0/34	ACTIVE
35	PC-35	FRANCA	VLAN 40	192.168.40.16	FastEthernet0/35	ACTIVE
36	PC-36	FRANCA	VLAN 40	192.168.40.17	FastEthernet0/36	ACTIVE
37	PC-37	FRANCA	VLAN 40	192.168.40.18	FastEthernet0/37	ACTIVE
38	PC-38	FRANCA	VLAN 40	192.168.40.19	FastEthernet0/38	ACTIVE
39	Switch F-A	CS & RECEPTION	****	****	FastEthernet0/39	ACTIVE
40	Switch F-B	CS & RECEPTION	****	****	FastEthernet0/40	ACTIVE
41	Switch-U	CS & RECEPTION	****	****	FastEthernet0/41	ACTIVE
42	Switch-U	CS & RECEPTION	****	****	FastEthernet0/42	ACTIVE
43	Switch-U	CS & RECEPTION	****	****	FastEthernet0/43	ACTIVE
44	Switch-U	CS & RECEPTION	****	****	FastEthernet0/44	ACTIVE
45	Switch-U	CS & RECEPTION	****	****	FastEthernet0/45	ACTIVE
46	Switch-U	CS & RECEPTION	****	****	FastEthernet0/46	ACTIVE
47	Switch-U	CS & RECEPTION	****	****	FastEthernet0/47	ACTIVE
48	Switch-U	CS & RECEPTION	****	****	FastEthernet0/48	ACTIVE
49	Switch-U	CS & RECEPTION	****	****	FastEthernet0/49	ACTIVE
50	Switch-U	CS & RECEPTION	****	****	FastEthernet0/50	ACTIVE
51	Switch-U	CS & RECEPTION	****	****	FastEthernet0/51	ACTIVE
52	Switch-U	CS & RECEPTION	****	****	FastEthernet0/52	ACTIVE
53	Switch-U	CS & RECEPTION	****	****	FastEthernet0/53	ACTIVE
54	Switch-U	CS & RECEPTION	****	****	FastEthernet0/54	ACTIVE
55	Switch-U	CS & RECEPTION	****	****	FastEthernet0/55	ACTIVE
56	Switch-U	CS & RECEPTION	****	****	FastEthernet0/56	ACTIVE
57	Switch-U	CS & RECEPTION	****	****	FastEthernet0/57	ACTIVE
58	Switch-U	CS & RECEPTION	****	****	FastEthernet0/58	ACTIVE
59	Switch-U	CS & RECEPTION	****	****	FastEthernet0/59	ACTIVE
60	Switch-U	CS & RECEPTION	****	****	FastEthernet0/60	ACTIVE
61	Switch-U	CS & RECEPTION	****	****	FastEthernet0/61	ACTIVE
62	Switch-U	CS & RECEPTION	****	****	FastEthernet0/62	ACTIVE
63	Switch-U	CS & RECEPTION	****	****	FastEthernet0/63	ACTIVE
64	Switch-U	CS & RECEPTION	****	****	FastEthernet0/64	ACTIVE
65	Switch-U	CS & RECEPTION	****	****	FastEthernet0/65	ACTIVE
66	Switch-U	CS & RECEPTION	****	****	FastEthernet0/66	ACTIVE
67	Switch-U	CS & RECEPTION	****	****	FastEthernet0/67	ACTIVE
68	Switch-U	CS & RECEPTION	****	****	FastEthernet0/68	ACTIVE
69	Switch-U	CS & RECEPTION	****	****	FastEthernet0/69	ACTIVE
70	Switch-U	CS & RECEPTION	****	****	FastEthernet0/70	ACTIVE
71	Switch-U	CS & RECEPTION	****	****	FastEthernet0/71	ACTIVE
72	Switch-U	CS & RECEPTION	****	****	FastEthernet0/72	ACTIVE
73	Switch-U	CS & RECEPTION	****	****	FastEthernet0/73	ACTIVE
74	Switch-U	CS & RECEPTION	****	****	FastEthernet0/74	ACTIVE
75	Switch-U	CS & RECEPTION	****	****	FastEthernet0/75	ACTIVE
76	Switch-U	CS & RECEPTION	****	****	FastEthernet0/76	ACTIVE
77	Switch-U	CS & RECEPTION	****	****	FastEthernet0/77	ACTIVE
78	Switch-U	CS & RECEPTION	****	****	FastEthernet0/78	ACTIVE
79	Switch-U	CS & RECEPTION	****	****	FastEthernet0/79	ACTIVE
80	Switch-U	CS & RECEPTION	****	****	FastEthernet0/80	ACTIVE
81	Switch-U	CS & RECEPTION	****	****	FastEthernet0/81	ACTIVE
82	Switch-U	CS & RECEPTION	****	****	FastEthernet0/82	ACTIVE
83	Switch-U	CS & RECEPTION	****	****	FastEthernet0/83	ACTIVE
84	Switch-U	CS & RECEPTION	****	****	FastEthernet0/84	ACTIVE
85	Switch-U	CS & RECEPTION	****	****	FastEthernet0/85	ACTIVE
86	Switch-U	CS & RECEPTION	****	****	FastEthernet0/86	ACTIVE
87	Switch-U	CS & RECEPTION	****	****	FastEthernet0/87	ACTIVE
88	Switch-U	CS & RECEPTION	****	****	FastEthernet0/88	ACTIVE
89	Switch-U	CS & RECEPTION	****	****	FastEthernet0/89	ACTIVE
90	Switch-U	CS & RECEPTION	****	****	FastEthernet0/90	ACTIVE
91	Switch-U	CS & RECEPTION	****	****	FastEthernet0/91	ACTIVE
92	Switch-U	CS & RECEPTION	****	****	FastEthernet0/92	ACTIVE
93	Switch-U	CS & RECEPTION	****	****	FastEthernet0/93	ACTIVE
94	Switch-U	CS & RECEPTION	****	****	FastEthernet0/94	ACTIVE
95	Switch-U	CS & RECEPTION	****	****	FastEthernet0/95	ACTIVE
96	Switch-U	CS & RECEPTION	****	****	FastEthernet0/96	ACTIVE
97	Switch-U	CS & RECEPTION	****	****	FastEthernet0/97	ACTIVE
98	Switch-U	CS & RECEPTION	****	****	FastEthernet0/98	ACTIVE
99	Switch-U	CS & RECEPTION	****	****	FastEthernet0/99	ACTIVE
100	Switch-U	CS & RECEPTION	****	****	FastEthernet0/100	ACTIVE

3. CONFIGURATION VERIFICATION & SECURITY STATUS

FLOOR 1 - CS & RECEPTION & FINANCE INFRASTRUCTURE					
NO	Device	Verification Parameter	Status	CLI Output	Evidence File Name
1	Switch F-CS-BT	VLAN ACCESS TRUNK ACCESS	ACTIVE	Interface VLAN 10 Interface VLAN 20 Interface VLAN 30	F.1-P.1
2	Router-U	IP DHCP POOL ACCESS	ACTIVE	IP DHCP POOL VLAN 10 IP DHCP POOL VLAN 20 IP DHCP POOL VLAN 30 IP DHCP POOL VLAN 40 IP DHCP POOL VLAN 50	F.1-P.2
3	Switch F-CS-BT	Enable Password Enable Password Why line Password Encryption Password Banner ACTIO	ACTIVE	enable secret enable password enable password enable password enable password	F.1-P.3
4	Switch F-CS-BT	Port Security	ACTIVE	Port security PoE/9-18	F.1-P.4

FLOOR 2 - IT ENGINEER & MANAGER INFRASTRUCTURE					
NO	Device	Verification Parameter	Status	CLI Output	Evidence File Name
1	Switch IT-MA	VLAN ACCESS TRUNK ACCESS	ACTIVE	Interface VLAN 40 Interface VLAN 50 Interface VLAN 60	F.2-P.1
3	Switch IT-MA	Enable Password Why line Password Encryption Password Banner ACTIO	ACTIVE	enable secret enable password enable password enable password enable password	F.2-P.2

4. BILL OF MATERIALS (BoQ) & INFRASTRUCTURE BUDGETING

NO	Device Name	Quantity Needed	Device Model	Unit Price	Total Price
1	PC	38	Intel Core i5-4570/4GB	1.500.000	57.000.000
2	SWITCH	2	Mikrotik CRS326-24G-2S-16G-2S	3.360.000	6.720.000
3	ROUTER	1	Mikrotik RB3009-12G-S	1.500.000	1.500.000
4	ROUTER	1	Mikrotik CRS326-24G-2S-16G-2S	3.360.000	3.360.000
Total Price					68.680.000

5. ENTERPRISE NETWORK DASHBOARD: VLAN & COST ANALYSIS

